AI-Use Disclosure - Quantum Oracle Forensics

1. AI Assistance Declared

I used Claude (Anthropic) via the claude.ai chat interface as an AI assistant during the hackathon. Event-specific formulation and code were developed during the event; no code, strategies, or oracle solutions from before the event were reused.

2. What the Assistant Produced

- **Initial Strategy & Code:** Designed the initial sequential Bayesian strategy (Fourier sampling, density test, Simon posterior) and wrote the first drafts of `solver.py`, `oracle_sim.py`, and the evaluation harness (`evaluate.py`, `baselines.py`).
- **Initial Testing:** Ran the initial 30-oracle/family evaluations in its sandbox and drafted the first version of the technical report.
- **Hyperparameter Tuning:** Suggested initial thresholds and the Grover iteration count $k = 8$ based on early sandbox simulations.

3. Independent Verification and Refinement by the Team

- **Code Review:** I read and understood the main source files (`solver.py`, `oracle_sim.py`, `evaluate.py`) line-by-line to verify the Bayesian posterior updates, density test math ($P(y=0) = \cos^2(2j\theta)$), and Simon subspace enumeration.
- **Large-Scale Validation:** I independently executed `python evaluate.py` for 50, 100, and 150 oracles per family (totaling 2,640 tasks) on my local machine. I used these results to override the AI's initial small-sample (30-oracle) optimistic claims, ensuring the report reflects statistically stable numbers (98.8% nominal, 95.3% stress).
- **Spec Compliance:** I ran `python test_spec.py` locally and confirmed all public examples pass.
- **Editorial Control:** I edited the report to add explicit mathematical equations and the multi-sample-size validation table to demonstrate rigorous scientific methodology.

4. Limits of AI Assistance

- All AI-generated logic was treated as a hypothesis and strictly verified against the local noisy simulator and public spec examples before inclusion.
- No hard-coded answers, seed lookups, or reading of oracle tables were used. All decisions derive strictly from the `run()` query results.
- Other tools used: Qiskit 2.x, Qiskit Aer, numpy, scipy, and standard Markdown-to-PDF exporters.